

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \rightarrow PA$ (8.1).

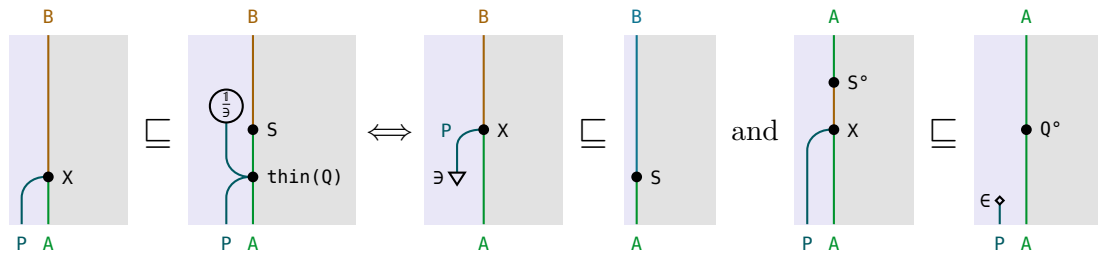
$ys \text{ thin}(Q) \ xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

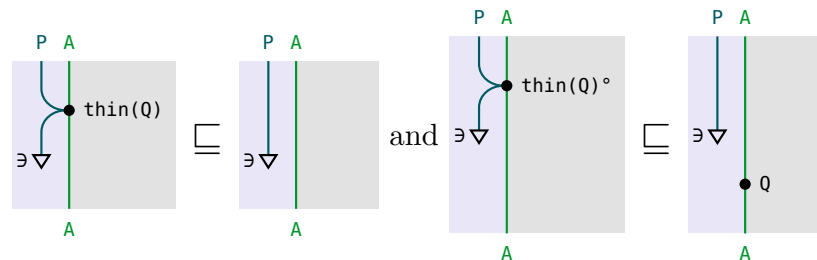
the law	what it says
1 $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \subseteq S \wedge S^\circ X \subseteq Q^\circ \epsilon$	S is the whole algebra: it produces every candidate that $\text{thin}(Q)$ thins. Everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
2 $\text{thin}(Q) \ni \subseteq \ni$	everything a thinning keeps was in the set
3 $\text{thin}(Q)^\circ \ni \subseteq \ni Q$	every element of the set has a Q -lower bound among the kept ones
4 $Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
5 $1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \subseteq Q \implies \text{thin}(Q) \text{ thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
6 $Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R) \text{ preorder}(Q)$ $\implies \text{est}(R) \subseteq \text{thin}(Q) \text{ est}(R)$	thin-introduction : thinning first cannot lose an R -minimum
7 $\text{thin}(Q) \supseteq \text{est}(Q) \frac{1}{\exists} \quad (8.2) - (8.2) \text{ est}(Q) \frac{1}{\exists} \subseteq \text{thin}(Q)$	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
8 $\frac{S}{\exists} \text{thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
9 $P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q) \quad (8.4) - (8.4)$ $P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q)$	thinning each member set is a thinning of the union

§8.1.1. $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \subseteq S \wedge S^\circ X \subseteq Q^\circ \epsilon$ — the definition of $\frac{S}{\exists} \text{thin}(Q)$

(8.1.1a)



(8.1.1b)



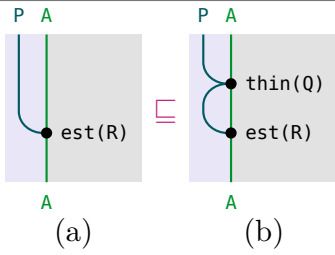
$X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \subseteq S \wedge S^\circ X \subseteq Q^\circ \epsilon$ at $X \triangleq \text{thin}(Q)$, $S \triangleq \ni$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\frac{\exists}{\exists} = \wedge(\ni) = 1$

§8.1.2. $Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \implies \text{est}(R) = \text{thin}(Q) \text{ est}(R)$ — thinning first, then taking the smallest, is the same as taking the smallest

$$\text{preorder}(Q) \Rightarrow \text{est}(R) \sqsubseteq \text{thin}(Q) \quad (8.1.2a)$$

est(R)

the $\sqsubseteq$ half: keeping everything is a
thinning — $1 \sqsubseteq Q$



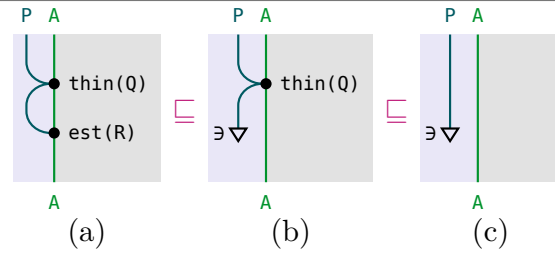
(a)	est(R)	$1 \sqsubseteq \text{thin}(Q) \text{ — } 1 \sqsubseteq Q$ $\Rightarrow 1 \sqsubseteq \text{thin}(Q),$ $Q \sqsubseteq Q$ $\Rightarrow \text{thin}(Q)$ $\text{thin}(Q) \sqsubseteq \text{thin}(Q)$
(b)	$\sqsubseteq \text{thin}(Q) \text{ est}(R)$	

$$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists \quad (8.1.2b)$$

the $\sqsubseteq$ half, first condition of the UP of

est at $X \triangleq \text{thin}(Q) \text{ est}(R) \text{ — } X \sqsubseteq$

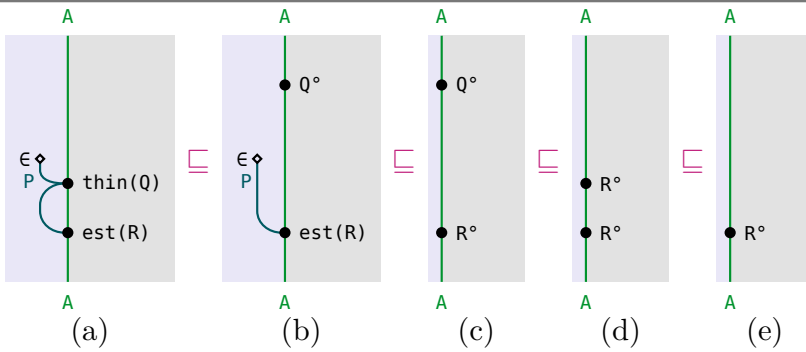
est(R) $\Leftrightarrow X \sqsubseteq \exists \wedge \in X \sqsubseteq R^\circ$



(a)	thin(Q) est(R)	$R \cap S \sqsubseteq R$
(b)	$\sqsubseteq \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(c)	$\sqsubseteq \exists$	

$$Q \sqsubseteq R \wedge \text{preorder}(R) \Rightarrow \in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ \quad (8.1.2c)$$

the $\sqsubseteq$ half, second condition



(a)	$\in \text{thin}(Q) \text{ est}(R)$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
(b)	$\sqsubseteq Q^\circ \in \text{est}(R)$	$\in \text{est}(R) \sqsubseteq R^\circ$
(c)	$\sqsubseteq Q^\circ R^\circ$	$Q \sqsubseteq R$
(d)	$\sqsubseteq R^\circ R^\circ$	
(e)	$\sqsubseteq R^\circ$	

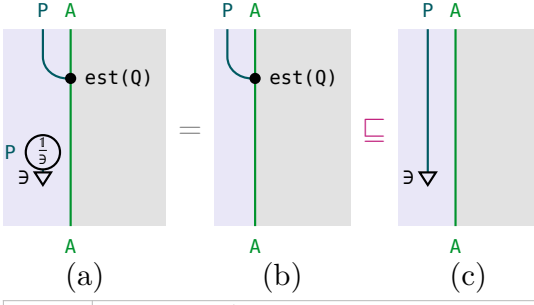
$$\mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq \mathbf{thin}(Q)$$

(8.2)

the singleton holding a Q -least member of a set is a thinning of that set

$$\mathbf{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

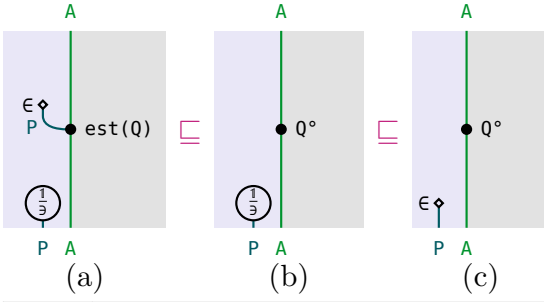
every member of the singleton is a member of the set — $\triangleq \mathbf{thin}, \ni/\ni$ half



(a)	$\mathbf{est}(Q) \stackrel{1}{\exists} \ni$	$\stackrel{1}{\exists} \ni = 1$
(b)	$= \mathbf{est}(Q)$	$R \cap S \sqsubseteq R$
(c)	$\sqsubseteq \ni$	

$$\in \mathbf{est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \in$$

the singleton's member stands in Q to every member of the set — $\triangleq \mathbf{thin}, \in/(Q^\circ \in)$ half

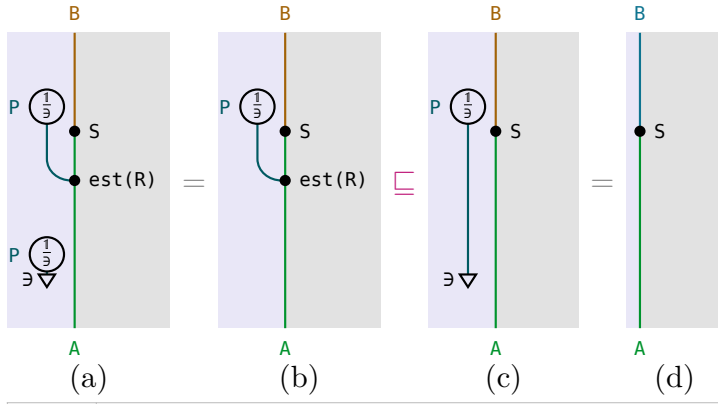


(a)	$\in \mathbf{est}(Q) \stackrel{1}{\exists}$	$\in \mathbf{est}(R) \sqsubseteq R^\circ$
(b)	$\sqsubseteq Q^\circ \stackrel{1}{\exists}$	$\stackrel{1}{\exists} \sqsubseteq \in$
(c)	$\sqsubseteq Q^\circ \in$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq \frac{S}{\exists} \text{thin}(Q) \quad (8.3)$$

$$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni \in S$$

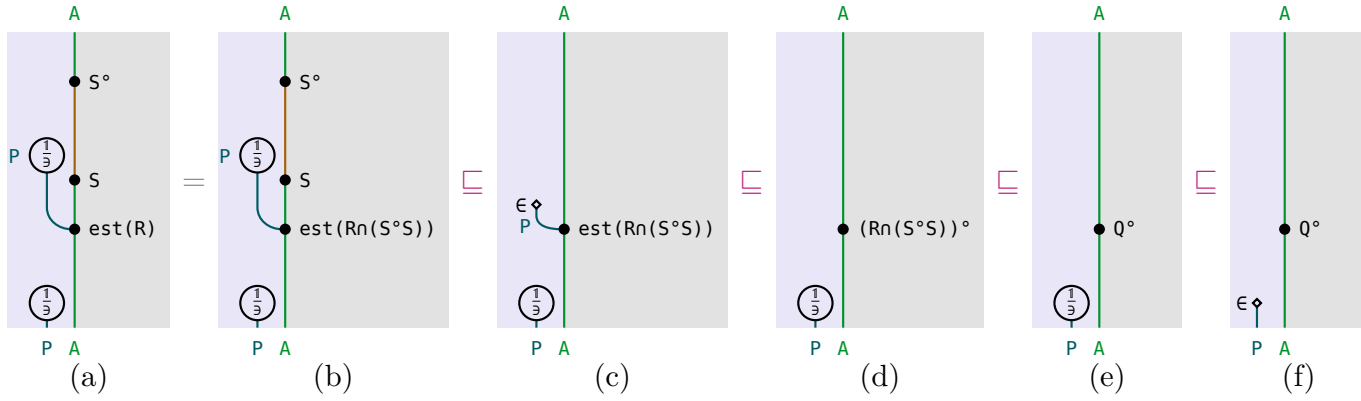
every member of the singleton is a value S returns — $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, first condition



(a)	$\frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \ni$	$\frac{1}{\exists} \ni = 1$
(b)	$= \frac{S}{\exists} \text{est}(R)$	$R \cap S \subseteq R$
(c)	$\sqsubseteq \frac{S}{\exists} \ni$	$\frac{R}{\exists} \ni = R$
(d)	$= S$	

$$R \cap (S^\circ S) \subseteq Q \Rightarrow S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq Q^\circ \in$$

the singleton's member stands in Q to every value S returns at the same argument — $X \subseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \in S \wedge S^\circ X \subseteq Q^\circ \in$, second condition



(a)	$S^\circ \frac{S}{\exists} \text{est}(R) \frac{1}{\exists}$	$\frac{S}{\exists} \text{est}(R \cap (S^\circ S)) = \frac{S}{\exists} \text{est}(R)$
(b)	$= S^\circ \frac{S}{\exists} \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$T^\circ \frac{1}{\exists} \subseteq \in$
(c)	$\sqsubseteq \in \text{est}(R \cap (S^\circ S)) \frac{1}{\exists}$	$\in \text{est}(R) \subseteq R^\circ$
(d)	$\sqsubseteq (R \cap (S^\circ S))^\circ \frac{1}{\exists}$	$R \cap (S^\circ S) \subseteq Q$
(e)	$\sqsubseteq Q^\circ \frac{1}{\exists}$	$\frac{1}{\exists} \subseteq \in$
(f)	$\sqsubseteq Q^\circ \in$	

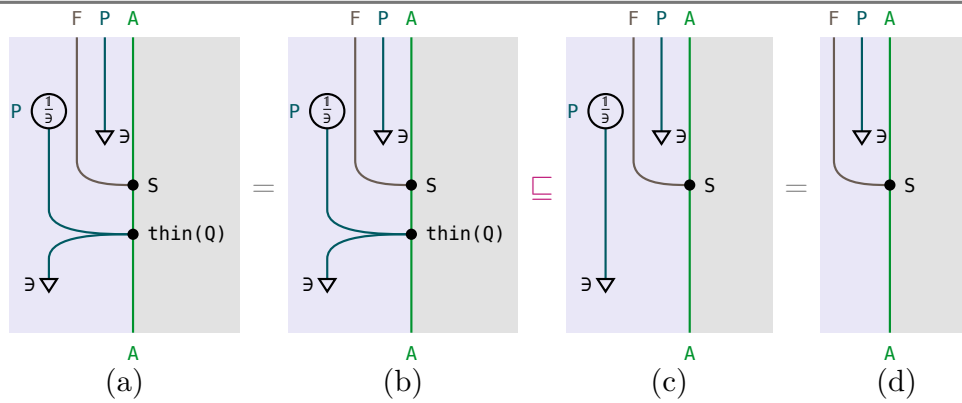
$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq (\frac{S}{\exists} \text{thin}(Q))$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \ni \sqsubseteq (S)$$

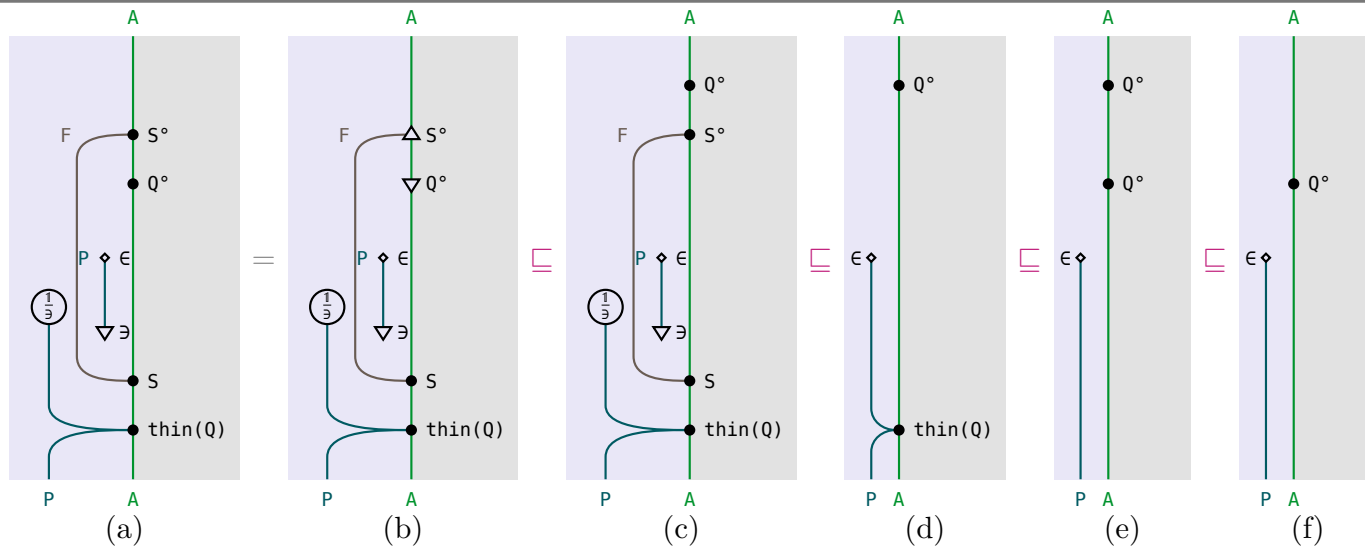
everything the thinning fold keeps is a value of the plain fold — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition, by fusion from the chain below



(a)	$\frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	
(b)	$= \frac{F(\exists) S}{\exists} \text{thin}(Q) \ni$	$\text{thin}(Q) \ni \sqsubseteq \exists$
(c)	$\sqsubseteq \frac{F(\exists) S}{\exists} \exists$	$\frac{R}{\exists} \exists = R$
(d)	$= F(\exists) S$	

$$\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q \implies (S)^\circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \sqsubseteq Q^\circ \in$$

every value of the plain fold is Q -above one the thinning fold keeps — $X \sqsubseteq \frac{S}{\exists} \text{thin}(Q) \iff X \ni \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition, by hylomorphism theorem from the chain below



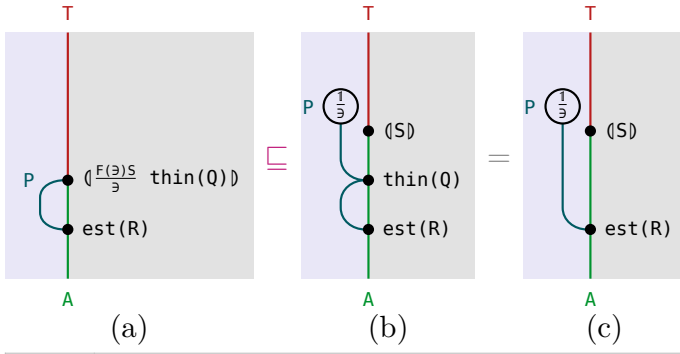
(a)	$S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	
(b)	$= S^\circ F(Q^\circ) F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$F(Q) S \sqsubseteq S Q$
(c)	$\sqsubseteq Q^\circ S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$\implies S^\circ F(Q^\circ) \sqsubseteq Q^\circ S^\circ$
(d)	$\sqsubseteq Q^\circ \in \text{thin}(Q)$	$S^\circ F(\epsilon) \frac{F(\exists) S}{\exists} \sqsubseteq \epsilon$
(e)	$\sqsubseteq Q^\circ Q^\circ \in$	$\epsilon \text{thin}(Q) \sqsubseteq Q^\circ \in$
(f)	$\sqsubseteq Q^\circ \in$	

(8.1.2d)

	step	type	example	what it does
1	$S:F(A)\rightarrow A$ is 8.1's algebra; example $F(X)=V+V\times X$, $S=[wrap,cons]$, candidate $a=cons(v,p)$			
2	S°	$A\rightarrow FA$	$cons(v,p) \mapsto r(v,p)$	takes a apart into the input of its last step
3	$F(Q^\circ\in)$	$FA\rightarrow F(PA)$	$r(v,p) \mapsto r(v,ps)$, some $p'\in ps$ with $p' \ Q \ p$	replaces each recursive component by a set holding a Q -no-worse one; v stays
4	$\frac{F(\exists)S}{\exists}$	$F(PA)\rightarrow PA$	$r(v,ps) \mapsto \{cons(v,t) \mid t\in ps\}$	picks one element from each set, combines by S , collects every result
5	$thin(Q)$	$PA\rightarrow PA$	$\{cons(v,t) \mid t\in ps\} \mapsto ys$	keeps a subset; every dropped element is Q -beaten by a kept one
6	$\sqsubseteq Q^\circ\in$	$A\rightarrow PA$	ys holds y with $y \ Q \ cons(v,p') \ Q \ cons(v,p) = a$	the thinned set still holds an element Q -no-worse than a (S monotonic on Q , Q transitive)

$$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) S \sqsubseteq S Q \Rightarrow (Q \xrightarrow{F(\exists) S} \text{thin}(Q)) \text{est}(R) \sqsubseteq \frac{(S)}{S} \text{est}(R) \quad \text{Corollary 8.1}$$

the thinning fold refines the optimisation problem itself



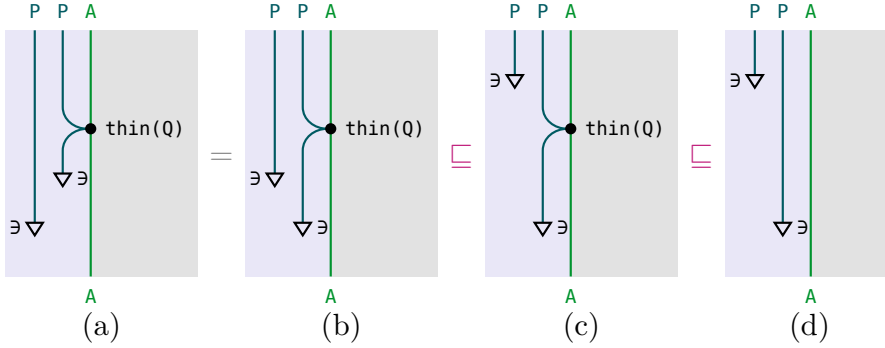
(a)	$(Q \xrightarrow{F(\exists) S} \text{thin}(Q)) \text{est}(R)$	Theorem 8.1 $\text{preorder}(Q) \wedge F(Q) S \sqsubseteq S Q$ $\Rightarrow (Q \xrightarrow{F(\exists) S} \text{thin}(Q)) \sqsubseteq \frac{(S)}{S} \text{thin}(Q)$
(b)	$\sqsubseteq \frac{(S)}{S} \text{thin}(Q) \text{est}(R)$	$Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R)$ $\Rightarrow \text{est}(R) = \text{thin}(Q) \text{est}(R)$
(c)	$= \frac{(S)}{S} \text{est}(R)$	

$$P(\text{thin}(Q)) \text{ union} \sqsubseteq \text{union thin}(Q) \quad (8.4)$$

thinning each member set, then taking the union, is a thinning of the union

$$P(\text{thin}(Q)) \text{ union} \exists \sqsubseteq \exists \exists$$

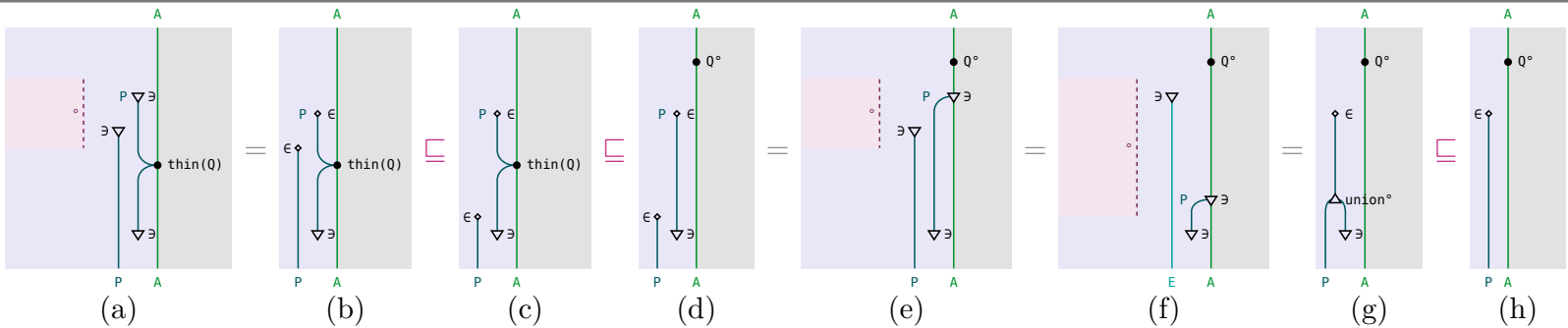
every member of the result is a member of a member set — $X \sqsubseteq \frac{S}{S} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, first condition



(a)	$P(\text{thin}(Q)) \text{ union} \exists$	$\text{union} \exists = \exists \exists$
(b)	$= P(\text{thin}(Q)) \exists \exists$	$P(R) \exists \sqsubseteq \exists R$
(c)	$\sqsubseteq \exists \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$
(d)	$\sqsubseteq \exists \exists$	

$$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union} \sqsubseteq Q^\circ \in$$

every member of a member set has a Q -lower bound in the result — $X \sqsubseteq \frac{S}{S} \text{thin}(Q) \iff X \exists \sqsubseteq S \wedge S^\circ X \sqsubseteq Q^\circ \in$, second condition



(a)	$(\exists \exists)^\circ P(\text{thin}(Q)) \text{ union}$	$(R S)^\circ = S^\circ R^\circ$
(b)	$= \in \in P(\text{thin}(Q)) \text{ union}$	$\in P(R) \sqsubseteq R \in$
(c)	$\sqsubseteq \in \text{thin}(Q) \in \text{union}$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$
(d)	$\sqsubseteq Q^\circ \in \in \text{union}$	$(R S)^\circ = S^\circ R^\circ$
(e)	$= Q^\circ (\exists \exists)^\circ \text{union}$	$\text{union} \exists = \exists \exists$
(f)	$= Q^\circ (\text{union} \exists)^\circ \text{union}$	$(R S)^\circ = S^\circ R^\circ$
(g)	$= Q^\circ \in \text{union}^\circ \text{union}$	$\text{union} = E(\exists)$
(h)	$\sqsubseteq Q^\circ \in$	

§8.2. Paths in a layered network

(8.2a)

	name	type	definition	meaning
1	F	$\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$	$F(B, X) = B + (B \times X)$	a path is either one vertex a , or a vertex a followed by a path xs
2	α	$F(A, LA) \rightarrow LA$	$\alpha = [\text{wrap}, \text{cons}]$	build a path: wrap one vertex as $[a]$, or put a in front of xs
3	wrapz	$A \rightarrow LA \times \mathbb{N}$	$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle, a \mapsto ([a], 0)$	a path of one vertex, no edge, cost 0
4	consw	$A \times (LA \times \mathbb{N}) \rightarrow LA \times \mathbb{N}$	$(a, (xs, n)) \mapsto ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$	put a in front of the path; the cost adds the weight of the new edge $a \rightarrow \text{head}(xs)$
5	cost	$LA \rightarrow \mathbb{N}$	$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$	the cost of $[a_0, \dots, a_n]$ is $\text{wt}(a_0, a_1) + \dots + \text{wt}(a_{n-1}, a_n)$
6	R	$LA \rightarrow LA$	$R \triangleq \text{cost} \leq \text{cost}^\circ$	xs is cheaper than ys
7	L(R)	$LA \rightarrow LB$	$\langle F(\exists, 1) \alpha \rangle = L(\exists)$	$L(\exists)$ relates a list of layers to every path that chooses one vertex from each layer
8	minpath	$L(PA) \rightarrow LA$	$\text{minpath} \sqsubseteq \frac{L(\exists)}{\exists} \text{est}(R)$	the input $[x_0, \dots, x_n] : L(PA)$ is a list of layers, each a set of vertices; a path chooses one vertex from each layer, and minpath returns a cheapest one
9	Q	$LA \rightarrow LA$	$Q \triangleq R \cap (\text{head} \text{ head}^\circ)$	like R , and the two paths also start at the same vertex: $\text{head}(xs) = \text{head}(ys)$
10	l	$B \rightarrow B + C$	$a \mapsto l(a)$	mark a value as the left alternative
11	r	$C \rightarrow B + C$	$b \mapsto r(b)$	mark a value as the right alternative
12	cpl	$PB \times A \rightarrow P(B \times A)$	$\text{cpl} \triangleq \text{cp}(1 \times A)$	pair every element of the set with the value beside it
13	cpr	$A \times PB \rightarrow P(A \times B)$	$\text{cpr} \triangleq \text{cp}(A \times 1)$	pair the value beside the set with every element of the set
14	$\frac{F(\exists, 1)}{\exists}$	$F(PB, X) \rightarrow PF(B, X)$	$\frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$	cpl transpose: from layer x and a set of paths ps , choose a vertex $a \in x$
15	$\frac{F(1, \exists)}{\exists}$	$F(B, PX) \rightarrow PF(B, X)$	$\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} P(l), \text{cpr } P(r)]$	cpr transpose: from a vertex a and a set of paths ps , choose a tail $xs \in ps$
16	step	$A \times P(LA) \rightarrow LA$	$\text{step} \triangleq \text{cpr } P(\text{cons}) \text{est}(R)$	put a in front of every path in ps and keep a cheapest one

(8.2b)

	the law	what it says
1	$F(\exists, Q) \alpha \sqsubseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
2	$S \text{ head} \sqsubseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \sqsubseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

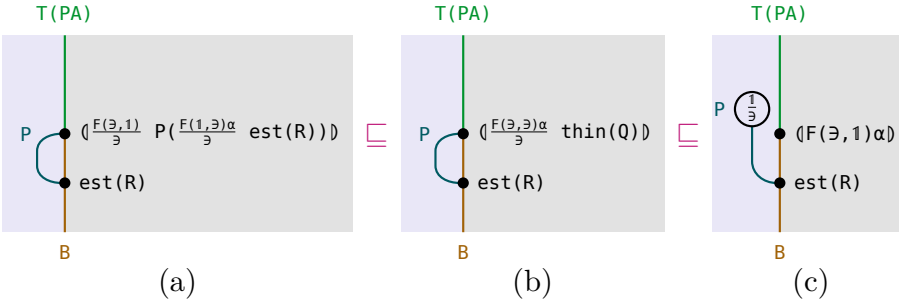
(8.2a)

$$Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(1, Q) F(\exists, 1) \alpha \subseteq F(\exists, 1) \alpha \wedge Q \wedge R \wedge ((F(1, \exists) \alpha)^\circ F(1, \exists) \alpha) \quad (8.2c)$$

$$\subseteq Q \Rightarrow \left(\frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \right) \text{est}(R) \subseteq \left(\frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$$

a least-cost path in a layered network, as a fold over the layers

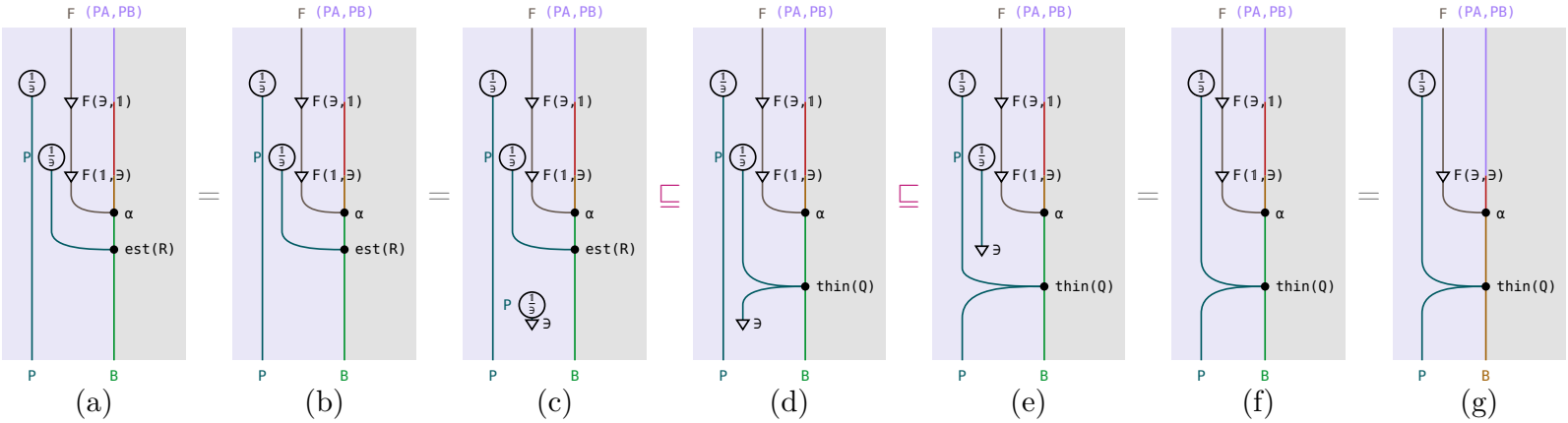
- Why it is needed: $\text{est}(R)$ keeps one cheapest w of the set $\Lambda(S)(x)$; $\text{thin}(Q)$ needs every dropped z to be Q -beaten by a kept one. For $\{w\}$ alone to be a thinning we need $w \leq_Q z$ for every z in $\Lambda(S)(x)$.
- We have $w \leq_R z$ (w is cheapest) and $w \leq_{(S^\circ S)} z$ (both are built by S from the same argument x); the hypothesis turns the two into $w \leq_Q z$: among candidates from one argument, cheaper already means Q -better.



(a)	$\left(\frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \right) \text{est}(R)$	(8.2d) $R \wedge (S^\circ S) \subseteq Q$ $\Rightarrow \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \subseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$
(b)	$\subseteq \left(\frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \right) \text{est}(R)$	$Q \subseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(1, Q)$ $F(\exists, 1) \alpha \subseteq F(\exists, 1) \alpha \wedge Q$
(c)	$\subseteq \left(\frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$	$\Rightarrow \left(\frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \right) \text{est}(R) \subseteq \left(\frac{F(\exists, 1) \alpha}{\exists} \right) \text{est}(R)$

$$S \triangleq F(1, \exists) \alpha, R \wedge (S^\circ S) \subseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \subseteq \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases



(a)	$\frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right)$	
(b)	$= \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right)$	$P \left(\frac{1}{\exists} \right) E(\exists) = 1$
(c)	$= \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \text{est}(R) \right) \frac{1}{\exists} E(\exists)$	(8.3) $R \wedge (S^\circ S) \subseteq Q$ $\Rightarrow \frac{S}{\exists} \text{est}(R) \frac{1}{\exists} \subseteq \frac{S}{\exists} \text{thin}(Q)$
(d)	$\subseteq \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \right) P(\text{thin}(Q)) E(\exists)$	(8.4) $P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q)$
(e)	$\subseteq \frac{F(\exists, 1)}{\exists} P \left(\frac{F(1, \exists) \alpha}{\exists} \right) E(\exists) \text{thin}(Q)$	$\frac{V S}{\exists} = \frac{V}{\exists} P \left(\frac{S}{\exists} \right) \text{ union}$
(f)	$= \frac{F(\exists, 1) F(1, \exists) \alpha}{\exists} \text{thin}(Q)$	$F(R, 1) F(1, S) = F(R, S)$
(g)	$= \frac{F(\exists, \exists) \alpha}{\exists} \text{thin}(Q)$	

$$= \text{cp}(F) P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}]$$

at the layered network $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}])$: $\frac{F(1, \exists)}{\exists} P(\alpha) \text{est}(R) = [\text{wrap}, \text{step}] - \text{step} \triangleq \text{cpr}$
 $P(\text{cons}) \text{est}(R)$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} - \frac{F(\exists, 1)}{\exists} = [P(l), \text{cpl } P(r)]$$

$$\begin{array}{ccccc} F(A, X) & = & A & + & A \times X \\ & & | & & | \\ & & R & & S \\ & & \downarrow & & \downarrow \\ F(A', X') & = & A' & + & A' \times X' \end{array}$$

(8.2e)

u $F(R, S)$ v holds iff one of:

- u, v are both inl and $d \leq_R d'$: the leaf moves by R;
- u, v are both inr and $p_1 \leq_R q_1 \wedge p_2 \leq_S q_2$: the vertex moves by R, the rest by S.
- An inl never relates to an inr : $R + S \triangleq [R \text{ inl}, S \text{ inr}]$ returns to the summand it came from, and $F(1) = 1$ allows no crossing.
- $F(\exists, \exists)$ relates $\text{inr}(\{a, b\}, \{p, q\})$ to $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$, and $\text{inl}(\{a, b\})$ to $\text{inl}(a), \text{inl}(b)$.

§8.3. Implementing thin

(8.3a)

	name	type	definition	meaning
1	setify	$[A] \rightarrow PA$	$(\text{setify } \exists)(x, a) \iff a \in x$	the set of a list's elements
2	cup	$PA \times PA \rightarrow PA$	$\text{cup} \triangleq \frac{(\pi_1 \exists) \cup (\pi_2 \exists)}{\exists}$	the union of two sets
3	cp(F)	$F(PA) \rightarrow PFA$	$\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$	every F-shaped value whose parts are chosen from the given sets
4	sort($\leq$)	$PA \rightarrow LA$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$	read a set back as one of its $\leq$ -ordered listings
5	bump(Q)	$A \times [A] \rightarrow [A]$	$(a, []) \mapsto [a], \text{bump}(Q)((a, [b] \# xs), ys) \iff$ $Q(a, b) \wedge ys = [a] \# xs \vee Q(b, a) \wedge ys = [b] \# xs \vee$ $\neg Q(a, b) \wedge \neg Q(b, a) \wedge ys = [a] \# [b] \# xs$	put a in front of a thinned list, dropping whichever of a and the old head the other beats
6	thinlist(Q)	$[A] \rightarrow [A]$	$\text{thinlist}(Q) = \emptyset[\text{nil}, \text{bump}(Q)]\emptyset$	thin a list in one pass, bumping each element into the thinned rest
7			$\text{IsThinlist}(Q, \text{thinlist}) \iff \text{thinlist} \sqsubseteq \text{subseq} \wedge$ $\text{thinlist setify} \sqsubseteq \text{setify thin}(Q)$	what a list implementation of thin(Q) must do: only drop elements, and thin the set
8	minlist(Q)	$[A] \rightarrow A$	$\text{minlist}(Q) \triangleq \text{setify est}(Q)$	a Q-least element of the list

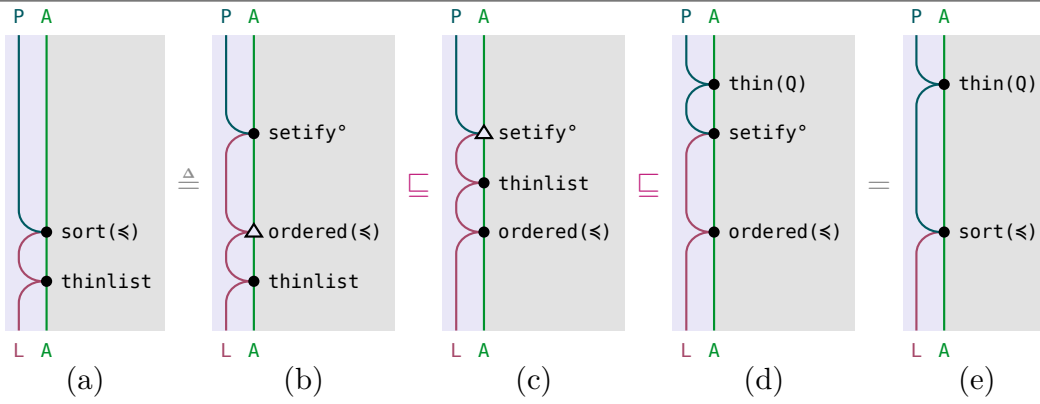
(8.3b)

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; $\leq$ a connected preorder with both f_1, f_2 monotonic on $\leq$; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$; $\text{listcp} : F(L) \rightarrow LF$.

(8.3c)

	the law	what it says
1	$\text{preorder}(Q) \wedge \text{connected}(Q) \implies \text{thinlist}(Q)$ $([a] \# xs, ys) \iff (\exists w. \text{minlist}(Q)([a] \# xs, w) \wedge ys = [w])$ (8.5)	what thinning should come to when it can: one element
2	$\text{preorder}(Q) \implies \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q)$ $\text{sort}(\leq)$ (8.6) — (8.6) $\text{Map}(\text{setify}) \wedge$ $\text{Coreflexive}(\text{ordered}(\leq)) \wedge \text{thinlist} \sqsubseteq \text{subseq} \wedge$ $\text{ordered}(\leq) \text{ subseq} \sqsubseteq \text{subseq ordered}(\leq) \wedge \text{thinlist}$ $\text{setify} \sqsubseteq \text{setify thin}(Q)$ $\implies \text{sort}(\leq) \text{ thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
3	$\text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R)$ (8.7)	a minimum of the sorted list is a minimum of the set
4	$\text{sort}(g \leq g^\circ) \text{ list}(g) \sqsubseteq P(g) \text{ sort}(\leq)$ (8.8)	shunt a function through a sort
5	$\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ (8.9)	filtering a sorted list sorts the restricted set
6	$\text{preorder}(\leq) \wedge \text{connected}(\leq) \implies (\text{sort}(\leq) \times \text{sort}(\leq))$ $\text{merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)	merging two sorted lists sorts their union
7	$F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$ (8.11), $FX = L + E \times X$	listcp is the list implementation of the cartesian product $\text{cp}(F)$

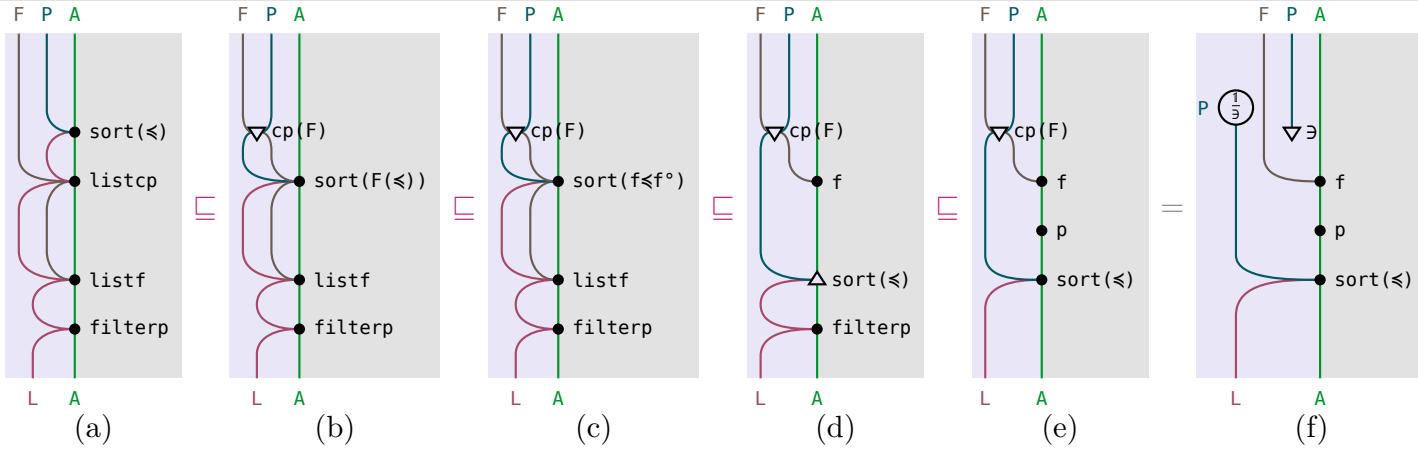
Map(setify) $\wedge$ Coreflexive(ordered($\leq$)) $\wedge$ thinlist $\sqsubseteq$ subseq $\wedge$ ordered($\leq$) subseq $\sqsubseteq$ subseq ordered($\leq$) $\wedge$ thinlist setify $\sqsubseteq$ setify thin(Q) $\Rightarrow$ sort($\leq$) thinlist $\sqsubseteq$ thin(Q) sort($\leq$) (8.6)
 a thinning of the sorted list lists a thinning of the set — $\leq$ a connected preorder, thinlist(Q) $\sqsubseteq$ subseq.



(a)	sort($\leq$) thinlist	sort($\leq$) $\triangleq$ setify $^\circ$ ordered($\leq$)
(b)	$\triangleq$ setify $^\circ$ ordered($\leq$) thinlist	Coreflexive(ordered($\leq$)) $\wedge$ g $\sqsubseteq$ subseq $\wedge$ ordered($\leq$) subseq $\sqsubseteq$ subseq ordered($\leq$) $\Rightarrow$ setify $^\circ$ ordered($\leq$) g $\sqsubseteq$ setify $^\circ$ g ordered($\leq$)
(c)	$\sqsubseteq$ setify $^\circ$ thinlist ordered($\leq$)	Map(setify) $\wedge$ g setify $\sqsubseteq$ setify T $\Rightarrow$ setify $^\circ$ g ordered($\leq$) $\sqsubseteq$ T setify $^\circ$ ordered($\leq$)
(d)	$\sqsubseteq$ thin(Q) setify $^\circ$ ordered($\leq$)	
(e)	= thin(Q) sort($\leq$)	

Map(f) $\wedge$ ($\forall X. \forall Y. X \sqsubseteq Y \Rightarrow$ sort(X) $\sqsubseteq$ sort(Y)) $\wedge$ F($\leq$) f $\sqsubseteq$ f $\leq$ $\wedge$ sort(f $\leq$ f $^\circ$) listf $\sqsubseteq$ P(f) sort($\leq$) $\wedge$ sort($\leq$) filterp $\sqsubseteq$ E(p) sort($\leq$) $\wedge$ F(sort($\leq$)) listcp $\sqsubseteq$ cp(F) sort(F($\leq$)) Lemma 8.1
 $\Rightarrow$ F(sort($\leq$)) listcp listf filterp $\sqsubseteq$ $\frac{F(\exists) f p}{\exists}$ sort($\leq$)

one sorted list built from sorted arguments, instead of a set built and then sorted — f : FA $\rightarrow$ A monotonic on $\leq$, p coreflexive, F linear.



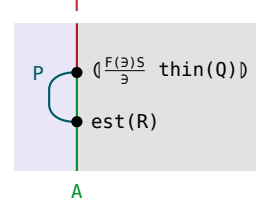
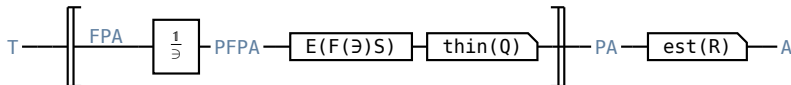
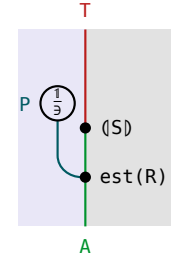
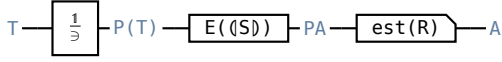
(a)	F(sort($\leq$)) listcp listf filterp	F(sort($\leq$)) listcp $\sqsubseteq$ cp(F) sort(F($\leq$))
(b)	$\sqsubseteq$ cp(F) sort(F($\leq$)) listf filterp	Map(f) $\Rightarrow$ F(R) f $\sqsubseteq$ f R $\Leftrightarrow$ F(R) $\sqsubseteq$ f R f $^\circ$
(c)	$\sqsubseteq$ cp(F) sort(f $\leq$ f $^\circ$) listf filterp	sort(f $\leq$ f $^\circ$) listf $\sqsubseteq$ P(f) sort($\leq$)
(d)	$\sqsubseteq$ cp(F) P(f) sort($\leq$) filterp	sort($\leq$) filterp $\sqsubseteq$ E(p) sort($\leq$) Map(f) $\Rightarrow$ P(f) = E(f)
(e)	$\sqsubseteq$ cp(F) E(f p) sort($\leq$)	$\frac{S}{\exists} E(R) = \frac{S R}{\exists}$
(f)	= $\frac{F(\exists) f p}{\exists}$ sort($\leq$)	

Theorem 8.2 $\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) \quad f_1 \ p_1 \sqsubseteq f_1 \ p_1 \ Q \wedge F(Q)$
 $f_2 \ p_2 \sqsubseteq f_2 \ p_2 \ Q \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sort}(X) \sqsubseteq \text{sort}(Y)) \wedge F(\leq) \quad f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \quad f_2 \sqsubseteq$
 $f_2 \leq \wedge \text{sort}(f_1 \leq f_1^\circ) \quad \text{listf}_1 \sqsubseteq P(f_1) \quad \text{sort}(\leq) \wedge \text{sort}(f_2 \leq f_2^\circ) \quad \text{listf}_2 \sqsubseteq P(f_2) \quad \text{sort}(\leq)$
 $\wedge \text{sort}(\leq) \quad \text{filterp}_1 \sqsubseteq E(p_1) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_2 \sqsubseteq E(p_2) \quad \text{sort}(\leq) \wedge F(\text{sort}(\leq))$
 $\text{listcp} \sqsubseteq \text{cp}(F) \quad \text{sort}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \quad \text{merge}(\leq) \sqsubseteq \text{cup} \quad \text{sort}(\leq) \wedge \text{sort}(\leq)$
 $\text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1 \quad \text{filterp}_1 \wedge$
 $g_2 = \text{listf}_2 \quad \text{filterp}_2 \wedge S = (f_1 \ p_1) \cup (f_2 \ p_2) \Rightarrow (\text{listcp} \ (g_1, g_2) \quad \text{merge}(\leq)$
 $\text{thinlist}(Q)) \quad \text{minlist}(R) \sqsubseteq \frac{(S)}{3} \text{est}(R)$

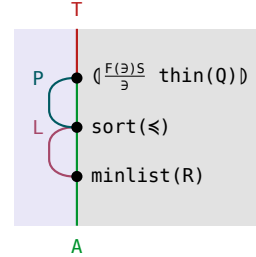
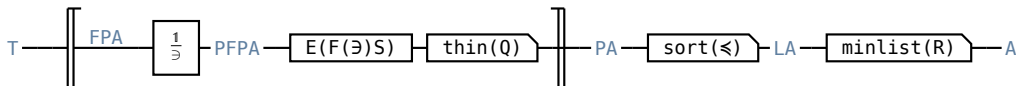
a fold on sorted lists of partial solutions, thinned at every step — at binary thinning data.

circuit

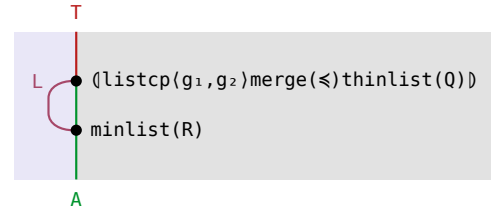
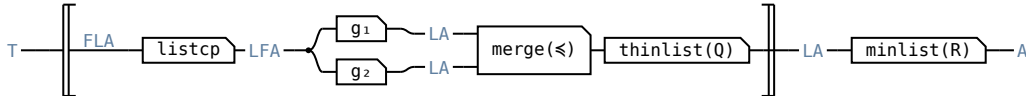
Hinze–Marsden



Corollary 8.1 $Q \sqsubseteq R \wedge \text{preorder}(Q) \wedge \text{preorder}(R) \wedge F(Q) \quad S \sqsubseteq S \quad Q$
 $\Rightarrow (\frac{F(3)S}{3} \text{thin}(Q)) \text{est}(R) \sqsubseteq \frac{(S)}{3} \text{est}(R)$ at $f_1 p_1$ and $f_2 p_2$ monotonic on Q —
 binary thinning data



$\text{sort}(\leq) \quad \text{minlist}(R) \sqsubseteq \text{est}(R)$ — $\text{sort}(\leq) \quad \text{minlist}(R) \sqsubseteq \text{est}(R)$ at its $Q \sqsubseteq R$



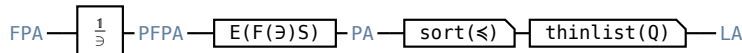
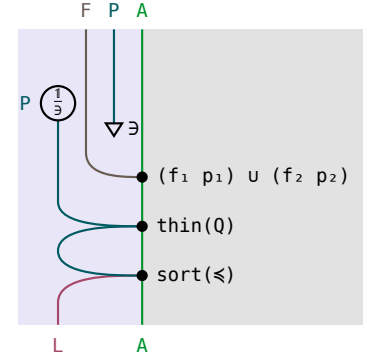
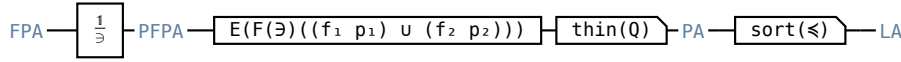
fusion at $\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sort}(X) \sqsubseteq \text{sort}(Y)) \wedge$
 $F(\leq) \quad f_1 \sqsubseteq f_1 \leq \wedge F(\leq) \quad f_2 \sqsubseteq f_2 \leq \wedge \text{sort}(f_1 \leq f_1^\circ) \quad \text{listf}_1 \sqsubseteq P(f_1)$
 $\text{sort}(\leq) \wedge \text{sort}(f_2 \leq f_2^\circ) \quad \text{listf}_2 \sqsubseteq P(f_2) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_1$
 $\sqsubseteq E(p_1) \quad \text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{filterp}_2 \sqsubseteq E(p_2) \quad \text{sort}(\leq) \wedge F(\text{sort}(\leq))$
 $\text{listcp} \sqsubseteq \text{cp}(F) \quad \text{sort}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \quad \text{merge}(\leq) \sqsubseteq \text{cup}$
 $\text{sort}(\leq) \wedge \text{sort}(\leq) \quad \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \quad \text{sort}(\leq)$
 $\Rightarrow F(\text{sort}(\leq)) \quad \text{listcp} \ (\text{listf}_1 \quad \text{filterp}_1, \text{listf}_2 \quad \text{filterp}_2) \quad \text{merge}(\leq)$
 $\text{thinlist}(Q) \sqsubseteq \frac{F(3) \ ((f_1 \ p_1) \cup (f_2 \ p_2))}{3} \text{thin}(Q) \quad \text{sort}(\leq)$

$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sort}(X) \sqsubseteq \text{sort}(Y)) \wedge F(\leq) f_1 \sqsubseteq f_1 \leq \wedge F(\leq) f_2 \sqsubseteq f_2 \leq \wedge \quad (8.3d) \\
& \text{sort}(f_1 \leq f_1^\circ) \text{listf}_1 \sqsubseteq P(f_1) \text{sort}(\leq) \wedge \text{sort}(f_2 \leq f_2^\circ) \text{listf}_2 \sqsubseteq P(f_2) \text{sort}(\leq) \wedge \text{sort}(\leq) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{sort}(\leq) \wedge \text{sort}(\leq) \text{filterp}_2 \sqsubseteq E(p_2) \text{sort}(\leq) \wedge F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \\
& \text{sort}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq) \wedge \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \text{listcp} (\text{listf}_1 \text{filterp}_1, \text{listf}_2 \text{filterp}_2) \text{merge}(\leq) \text{thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{thin}(Q) \text{sort}(\leq)
\end{aligned}$$

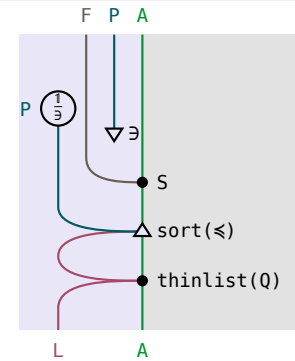
sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit

Hinze–Marsden



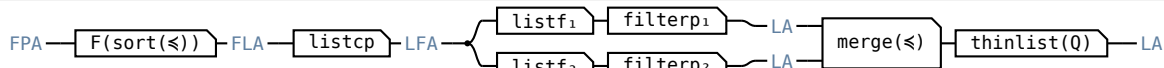
$$\begin{aligned}
& \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq) - \text{preorder}(Q) \\
& \Rightarrow \text{sort}(\leq) \text{thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{sort}(\leq)
\end{aligned}$$



$$\begin{aligned}
& \text{FPA} \rightarrow \left[\begin{array}{l} \text{[1/3]} \rightarrow \text{PFPA} \rightarrow \text{E(F(}\exists\text{) } f_1 p_1) \\ \text{[1/3]} \rightarrow \text{PFPA} \rightarrow \text{E(F(}\exists\text{) } f_2 p_2) \end{array} \right] \rightarrow \text{PA} \rightarrow \text{cup} \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left(\frac{F(\exists) f_1 p_1}{\exists}, \frac{F(\exists) f_2 p_2}{\exists} \right) \text{cup} \text{sort}(\leq) \text{thinlist}(Q) \\
& S = (f_1 p_1) \cup (f_2 p_2) - \text{binary thinning data, then } \triangleq \text{cup}
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow \left[\begin{array}{l} \text{[1/3]} \rightarrow \text{PFPA} \rightarrow \text{E(F(}\exists\text{) } f_1 p_1) \\ \text{[1/3]} \rightarrow \text{PFPA} \rightarrow \text{E(F(}\exists\text{) } f_2 p_2) \end{array} \right] \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left(\frac{F(\exists) f_1 p_1}{\exists}, \frac{F(\exists) f_2 p_2}{\exists} \right) (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \text{thinlist}(Q) \\
& (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq) - \text{preorder}(\leq) \wedge \text{connected}(\leq) \\
& \Rightarrow (\text{sort}(\leq) \times \text{sort}(\leq)) \text{merge}(\leq) \sqsubseteq \text{cup} \text{sort}(\leq)
\end{aligned}$$

$$\begin{aligned}
& \text{FPA} \rightarrow \left[\begin{array}{l} \text{[1/3]} \rightarrow \text{PFPA} \rightarrow \text{E(F(}\exists\text{) } f_1 p_1) \\ \text{[1/3]} \rightarrow \text{PFPA} \rightarrow \text{E(F(}\exists\text{) } f_2 p_2) \end{array} \right] \rightarrow \text{PA} \rightarrow \text{sort}(\leq) \rightarrow \text{LA} \rightarrow \text{merge}(\leq) \rightarrow \text{thinlist}(Q) \rightarrow \text{LA} \\
& = \left(\frac{F(\exists) f_1 p_1}{\exists} \text{sort}(\leq), \frac{F(\exists) f_2 p_2}{\exists} \text{sort}(\leq) \right) \text{merge}(\leq) \text{thinlist}(Q) \\
& \langle X, Y \rangle (\text{sort}(\leq) \times \text{sort}(\leq)) = \langle X \text{sort}(\leq), Y \text{sort}(\leq) \rangle - \text{absorption}
\end{aligned}$$



$$F(\text{sort}(\leq)) \text{listcp} (g_1, g_2) \text{merge}(\leq) \text{thinlist}(Q)$$

Lemma 8.1 $\text{Map}(f) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sort}(X) \sqsubseteq \text{sort}(Y)) \wedge F(\leq) f \sqsubseteq f \leq \wedge$

$$\begin{aligned}
& \text{sort}(f \leq f^\circ) \text{listf} \sqsubseteq P(f) \text{sort}(\leq) \wedge \text{sort}(\leq) \text{filterp} \sqsubseteq E(p) \text{sort}(\leq) \wedge \\
& F(\text{sort}(\leq)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(F(\leq)) \\
& \Rightarrow F(\text{sort}(\leq)) \text{listcp} \text{listf} \text{filterp} \sqsubseteq \frac{F(\exists) f p}{\exists} \text{sort}(\leq) \text{ at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then} \\
& X \langle g_1, g_2 \rangle \sqsubseteq \langle X g_1, X g_2 \rangle - X \langle R, S \rangle \sqsubseteq \langle X R, X S \rangle ; g_1 \triangleq \text{list}(f_1) \text{filter}(p_1) - \text{binary} \\
& \text{thinning data}
\end{aligned}$$

§8.4. The knapsack problem

(8.4a)

	name	type	definition	meaning
1	$\text{total}(f,x)$	Int	$\text{total}(f,x) \triangleq \text{sum}(\text{list}(f,x))$, $\text{total } f = \text{list}(f) \text{ sum}$	add up f over the items of the list
2	subseq	$[A] \rightarrow [A]$	$x \text{ subseq } ys \iff \text{subseq}(ys,x)$	every subsequence of the list
3	$[\text{nil}, \text{cons}]$	$F[\text{Item}] \rightarrow [\text{Item}]$	$[\text{nil}, \text{cons}] = [\text{nil}, \text{cons}]$	keep the item: the list constructor
4	$[\text{nil}, \pi_2]$	$\text{Unit} + \text{Item} \times [\text{Item}] \rightarrow [\text{Item}]$	$[\text{nil}, \pi_2](\text{l}(D)) = []$, $[\text{nil}, \pi_2](\text{r}(p)) = \pi_2(p)$, $[\text{nil}, \pi_2] = [\text{nil}, \pi_2]$	drop the item: keep the tail
5	$\text{within}(w)$	$[\text{Item}] \rightarrow [\text{Item}]$	$\text{within}(w)(x,y) \iff x = y \wedge \text{total}(\text{wt},x) \leq w$	the packings whose total weight is at most w
6	$([\text{nil}, \text{cons}] \text{ within}(w)) \cup [\text{nil}, \pi_2]$	$F[\text{Item}] \rightarrow [\text{Item}]$	$([\text{nil}, \text{cons}] \text{ within}(w)) \cup [\text{nil}, \pi_2] \triangleq$ $([\text{nil}, \text{cons}] \text{ within}(w)) \cup [\text{nil}, \pi_2]$, $([\text{nil}, \text{cons}] \text{ within}(w)) \cup [\text{nil}, \pi_2] = [\text{nil},$ $(\text{cons } \text{ within}(w)) \cup \pi_2]$	at each item, keep it if the packing still fits, or drop it
7	R	$[\text{Item}] \rightarrow [\text{Item}]$	$R = \text{total vol} \geq \text{total vol}^\circ$, $R(x,y) \iff \text{total}(\text{vol},y) \leq \text{total}(\text{vol},x)$	x is worth at least as much as y
8	$Q(\text{vol}, \text{wt})$	$[\text{Item}] \rightarrow [\text{Item}]$	$Q(\text{vol}, \text{wt}) = R \cap (\text{total wt} \leq \text{total wt}^\circ)$, $Q(\text{vol}, \text{wt})(x,y) \iff \text{total}(\text{vol},y) \leq \text{total}(\text{vol},x) \wedge \text{total}(\text{wt},x) \leq \text{total}(\text{wt},y)$	x is worth at least as much as y and weighs no more

(8.4b)

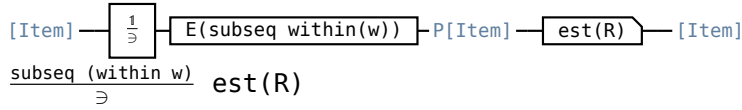
	the law	what it says
1	$(1 \times R) (\text{cons } (\text{within } w)) \sqsubseteq \text{cons } (\text{within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
2	$F(Q(\text{vol}, \text{wt})) [\text{nil}, \text{cons}] \text{ within}(w) \sqsubseteq [\text{nil}, \text{cons}] \text{ within}(w) Q(\text{vol}, \text{wt})$ $F(Q(\text{vol}, \text{wt})) [\text{nil}, \pi_2] \sqsubseteq [\text{nil}, \pi_2] Q(\text{vol}, \text{wt})$,	both halves are monotonic on Q once ties in value are broken by weight
3	$F(R) [\text{nil}, \text{cons}] \sqsubseteq [\text{nil}, \text{cons}] R$ $F(R) [\text{nil}, \pi_2] \sqsubseteq [\text{nil}, \pi_2] R$	both algebras are monotonic on R itself, so R is the sort order P

(8.4a)

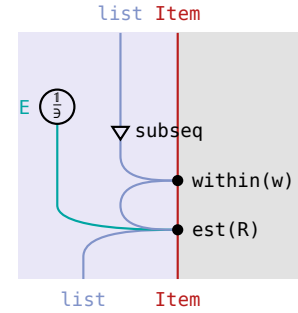
$$\begin{aligned}
& 0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sort}(X) \sqsubseteq \text{sort}(Y)) \wedge \text{sort}([\text{nil}, \text{cons}] R \\
& [\text{nil}, \text{cons}]^\circ) \text{listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{sort}(R) \wedge \text{sort}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{listf}_2 \sqsubseteq P([\text{nil}, \pi_2]) \\
& \text{sort}(R) \wedge \text{sort}(R) \text{filterp}_1 \sqsubseteq E(\text{within}(w)) \text{sort}(R) \wedge F(\text{sort}(R)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sort}(F(R)) \wedge \\
& (\text{sort}(R) \times \text{sort}(R)) \text{merge}(R) \sqsubseteq \text{cup} \text{sort}(R) \wedge \text{sort}(R) \text{thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt})) \text{sort}(R) \wedge \\
& \text{sort}(R) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow \underbrace{(\text{listcp} (\text{listf}_1 \text{filterp}_1, \text{listf}_2) \text{merge}(R) \text{thinlist}) \text{minlist}}_{\text{subseq within}(w)} \sqsubseteq \text{est}(R)
\end{aligned} \tag{8.4c}$$

the knapsack problem, as a fold that thins the packings kept at each item

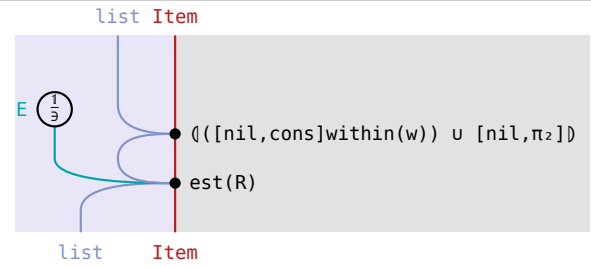
circuit



Hinze–Marsden



$$\begin{aligned}
& [\text{Item}] \xrightarrow{\boxed{\frac{1}{3}}} E(\underbrace{([\text{nil}, \text{cons}] \text{within}(w)) \cup [\text{nil}, \pi_2]}_{\text{fusion, weights non-negative}}) \text{P}[\text{Item}] \xrightarrow{\boxed{\text{est}(R)}} [\text{Item}] \\
& = \underbrace{([\text{nil}, \text{cons}] \text{within}(w)) \cup [\text{nil}, \pi_2]}_{\text{fusion, weights non-negative}} \text{est}(R)
\end{aligned}$$



$$\begin{aligned}
& [\text{Item}] \xrightarrow{\boxed{\frac{1}{3}}} \text{listcp} \text{list}(\text{F}[\text{Item}]) \xrightarrow{\boxed{\text{listf}_1 \text{filterp}_1 \text{listf}_2}} \text{list}[\text{Item}] \xrightarrow{\boxed{\text{merge}(R)}} \text{thinlist} \text{list}[\text{Item}] \xrightarrow{\boxed{\text{minlist}}} [\text{Item}] \\
& \sqsubseteq \text{binary thinning theorem, at } P \triangle R, F \text{ linear, } Q \text{ from } F(Q(\text{vol}, \text{wt})) \\
& [\text{nil}, \text{cons}] \text{within}(w) \sqsubseteq [\text{nil}, \text{cons}] \text{within}(w) \\
& Q(\text{vol}, \text{wt}), F(Q(\text{vol}, \text{wt})) [\text{nil}, \pi_2] \sqsubseteq [\text{nil}, \pi_2] Q(\text{vol}, \text{wt})
\end{aligned}$$



$$\begin{aligned}
& Q[\text{nil}, \text{cpr} \langle h_1, h_2 \rangle \text{merge} R \text{thinlist}(Q)] \text{minlist}(R) \\
& = \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{nil}), h_i] \text{---} [\text{nil}, \text{cons}] = \\
& [\text{nil}, \text{cons}], [\text{nil}, \pi_2](l(D)) = [], [\text{nil}, \pi_2](r(p)) = \\
& \pi_2(p); \text{minlist}(R) \text{ is head, packings coming out in descending value}
\end{aligned}$$

§8.5. The paragraph problem

(8.5a)

	name	type	definition	meaning
1	$\text{list}^+(\text{Word})$	Type	$\text{list}^+(\text{Word}) \triangleq \text{L}(\text{Word})$	a line is a non-empty list of words
2	$\text{list}^+(\text{list}^+ \text{ Word})$	Type	$\text{list}^+(\text{list}^+ \text{ Word}) \triangleq \text{L}(\text{list}^+(\text{Word}))$	a paragraph is a non-empty list of lines
3	$\text{new}(a,p)$	$\text{list}^+ (\text{list}^+ \text{ Word})$	$\text{new}(a,p) \triangleq [[a]]\#p$	start a new first line holding only the word a
4	$\text{glue}(a)$	$\text{list}^+ (\text{list}^+ \text{ Word}) \rightarrow \text{list}^+ (\text{list}^+ \text{ Word})$	$\text{glue}(a,[l]) = [[a]\#l], \text{glue}(a,[l]\#a_1) = [[a]\#l]\#a_1$	put the word a at the front of the first line
5	$[\text{wrap wrap,new}]$	$F (\text{list}^+ (\text{list}^+ \text{ Word})) \rightarrow \text{list}^+ (\text{list}^+ \text{ Word})$	$[\text{wrap wrap,new}] (\text{l}(a)) = [[a]], [\text{wrap wrap,new}] (r(q)) = \text{new}(\pi_1(q), \pi_2(q))$	one word is a one-line paragraph; each further word starts a new line
6	$[\text{wrap wrap,glue}]$	$F (\text{list}^+ (\text{list}^+ \text{ Word})) \rightarrow \text{list}^+ (\text{list}^+ \text{ Word})$	$[\text{wrap wrap,glue}] (\text{l}(a)) = [[a]], [\text{wrap wrap,glue}] (r(q)) = \text{glue}(\pi_1(q), \pi_2(q))$	one word is a one-line paragraph; each further word joins the first line
7	$[\text{wrap wrap,new} \cup \text{glue}]$	$F(\text{list}^+(\text{list}^+ \text{ Word})) \rightarrow \text{list}^+(\text{list}^+ \text{ Word})$	$[\text{wrap wrap,new} \cup \text{glue}] \triangleq [\text{wrap wrap,new}] \cup [\text{wrap wrap,glue}]$	each further word either starts a new line or joins the first line
8	partition	$\text{L}(\text{Word}) \rightarrow \text{list}^+(\text{list}^+ \text{ Word})$	$\text{partition} \triangleq ([\text{wrap wrap,new} \cup \text{glue}])$	every way of breaking the words into lines
9	width	$\text{list}^+ \text{ Word} \rightarrow \text{Int}$	$\text{width}[a] = \text{len}(a), \text{width}([a]\#x_1) = (\text{len}(a) + \text{width}(x_1)) + 1$	the lengths of the words plus one space between neighbours
10	head	$\text{list}^+ (\text{list}^+ \text{ Word}) \rightarrow \text{list}^+ \text{ Word}$	$\text{head}[l] = l, \text{head}([l]\#a) = l$	the first line of the paragraph
11	$\text{fits}(w)$	$P (\text{list}^+ (\text{list}^+ \text{ Word}))$	$\text{fits}(w,[l]) = \text{width}(l) \leq w, \text{fits}(w,[l]\#a) = \text{width}(l) \leq w \wedge \text{fits}(w,a)$	every line is at most w wide
12	$\text{fits}(w)$	$\text{list}^+(\text{list}^+ \text{ Word}) \rightarrow \text{list}^+(\text{list}^+ \text{ Word})$	$\text{fits}(w)(p,q) \iff p = q \wedge \text{fits}(w,p)$	the paragraphs whose every line is at most w wide
13	$\text{ok}(w)$	$\text{list}^+(\text{list}^+ \text{ Word}) \rightarrow \text{list}^+(\text{list}^+ \text{ Word})$	$\text{ok}(w)(p,q) \iff p = q \wedge \text{width}(\text{head}(p)) \leq w$	the paragraphs whose first line is at most w wide
14	$\text{sqr}(n)$	Int	$\text{sqr}(n) \triangleq nn$	n squared
15	$\text{waste}(w)$	$\text{list}^+ (\text{list}^+ \text{ Word}) \rightarrow \text{Int}$	$\text{waste}(w,[l]) = 0, \text{waste}(w,[l]\#a) = \text{sqr}(w - \text{width } l) + \text{waste}(w,a)$	the squared white space left on every line but

	name	type	definition	meaning
				the last, added up
16	R	$\text{list}^+(\text{list}^+ \text{ Word}) \rightarrow \text{list}^+(\text{list}^+ \text{ Word})$	$R = \text{waste } w \leq \text{waste } w^\circ, R(p,q) \iff \text{waste}(w,p) \leq \text{waste}(w,q)$	p wastes no more than q
17	$Q(\text{len},w)$	$\text{list}^+(\text{list}^+ \text{ Word}) \rightarrow \text{list}^+(\text{list}^+ \text{ Word})$	$Q(\text{len},w) = R \cap (\text{head } \text{head}^\circ), Q(\text{len},w)(p,q) \iff \text{waste}(w,p) \leq \text{waste}(w,q) \wedge \text{head}(p) = \text{head}(q)$	p wastes no more than q and has the same first line
18	start(a)	$[\text{list}^+ (\text{list}^+ \text{ Word})]$	$\text{start}(a) \triangleq [[a]]$	the one candidate for a single word: the paragraph of one line holding that word

(8.5b)

	the law	what it says
1	$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
2	$F(Q(\text{len},w)) [\text{wrap wrap,new}] \sqsubseteq [\text{wrap wrap,new}] Q(\text{len},w)$ $(\forall a. 0 \leq \text{len}(a)) \implies F(Q(\text{len},w)) [\text{wrap wrap,glue}] \text{ok}(w) \sqsubseteq [\text{wrap wrap,glue}] \text{ok}(w) Q(\text{len},w)$ cons monotonic on $\text{collect} \leq \text{collect}^\circ$.	both halves are monotonic on Q once ties in waste are broken by the first line
3	$\text{merge}(\tau) = \text{cat}$	τ needs no sorting at all
4	$F(\tau) [\text{wrap wrap,new}] \sqsubseteq [\text{wrap wrap,new}] \tau$ $F(\tau) [\text{wrap wrap,glue}] \sqsubseteq [\text{wrap wrap,glue}] \tau$	both algebras are monotonic on τ , so τ is the sort order P

§8.6. Bitonic tours

18

	name	type	definition	meaning
7	incost	$\text{Journey City} \rightarrow \text{Int}$	$\text{incost}(\text{wrap}(p)) = \text{tc}(\pi_2(p), \pi_1(p)),$ $\text{incost}([a]\#a_1) = \text{tc}(\text{head}(a_1), a) + \text{incost}(a_1)$	the cost of travelling the journey backwards
8	cost(t)	Int	$\text{cost}(t) \triangleq \text{outcost}(\pi_1(t)) + \text{incost}(\pi_2(t))$	the outward journey forwards plus the return journey backwards
9	start	$\text{City} \times \text{City} \rightarrow \text{Tour City}$	$\text{start}(a) \triangleq (\text{wrap}(a), \text{wrap}(a))$	the tour of the last two cities, both journeys being that one edge
10	dropl(a,t)	Tour City	$\text{dropl}(a,t) \triangleq (\text{replaceHead}(a, \pi_1(t)), [a]\# \pi_2(t))$	a replaces the first city of the outward journey and is put in front of the return
11	dropr(a,t)	Tour City	$\text{dropr}(a,t) \triangleq ([a]\# \pi_1(t), \text{replaceHead}(a, \pi_2(t)))$	a is put in front of the outward journey and replaces the first city of the return
12	[start,dropl]	$F(\text{dTour City}) \rightarrow \text{Tour City}$	$[\text{start}, \text{dropl}](l(p)) = \text{start}(p), [\text{start}, \text{dropl}](r(q)) = \text{dropl}(\pi_1(q), \pi_2(q))$	start at the last two cities, then add each city by dropl
13	[start,dropr]	$F(\text{dTour City}) \rightarrow \text{Tour City}$	$[\text{start}, \text{dropr}](l(p)) = \text{start}(p), [\text{start}, \text{dropr}](r(q)) = \text{dropr}(\pi_1(q), \pi_2(q))$	start at the last two cities, then add each city by dropr
14	[start,dropl u dropr]	$F(\text{dTour}(\text{City})) \rightarrow \text{dTour}(\text{City})$	$[\text{start}, \text{dropl u dropr}] \triangleq [\text{start}, \text{dropl}] \cup [\text{start}, \text{dropr}]$	each further city is added by dropl or by dropr
15	Q([start,dropl u dropr])	$\mu F \text{dTour}(\text{City})$	$Q([\text{start}, \text{dropl u dropr}]) \triangleq Q([\text{start}, \text{dropl u dropr}])$	every bitonic tour of the cities
16	R	$\text{dTour}(\text{City}) \rightarrow \text{dTour}(\text{City})$	$R = \text{cost} \leq \text{cost}^\circ, R(t, t') \iff \text{cost}(t) \leq \text{cost}(t')$	t costs no more than t'
17	next2(t)	$\text{City} \times \text{City}$	$\text{next2}(t) \triangleq (\text{next}(\pi_1(t)), \text{next}(\pi_2(t)))$	the second cities of both journeys
18	head2(t)	$\text{City} \times \text{City}$	$\text{head2}(t) \triangleq (\text{head}(\pi_1(t)), \text{head}(\pi_2(t)))$	the first cities of both journeys
19	Q	$\text{dTour}(\text{City}) \rightarrow \text{dTour}(\text{City})$	$Q = R \cap (\text{next2 next2}^\circ) \cap (\text{head2 head2}^\circ), Q(t, t') \iff \text{cost}(t) \leq \text{cost}(t') \wedge (\text{next}(\pi_1(t)) = \text{next}(\pi_1(t')) \wedge \text{next}(\pi_2(t)) = \text{next}(\pi_2(t'))) \wedge (\text{head}(\pi_1(t)) = \text{head}(\pi_1(t')) \wedge \text{head}(\pi_2(t)) = \text{head}(\pi_2(t')))$	t costs no more than t', and both journeys of t and t' share their first two cities

(8.6b)

	the law	what it says
1	$(1 \times R) \text{dropl} \sqsubseteq \text{dropl} \quad R \text{ FALSE}$ $(1 \times R) \text{dropr} \sqsubseteq \text{dropr} \quad R \text{ FALSE}$	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
2	$F(Q) [\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] \quad Q$ $F(Q) [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] \quad Q$	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input
3	$F(T) [\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] \quad T$ $F(T) [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] \quad T$	both algebras are monotonic on T, so T is the sort order P

$$\begin{aligned}
& (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sort}(X) \sqsubseteq \text{sort}(Y)) \wedge \text{sort}([\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}])^\circ \vdash g_1 \sqsubseteq \text{P}([\text{start}, \text{dropl}]) \text{ sort}(\tau) \wedge \text{sort}([\text{start}, \text{dropr}] \sqcup [\text{start}, \text{dropl}])^\circ \vdash g_2 \sqsubseteq \text{P}([\text{start}, \text{dropr}]) \text{ sort}(\tau) \\
& \wedge F(\text{sort}(\tau)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{ cat} \sqsubseteq \text{cup} \text{ sort}(\tau) \wedge \text{sort}(\tau) \\
& \text{thinlist} \sqsubseteq \text{thin}(Q) \text{ sort}(\tau) \wedge \text{sort}(\tau) \text{ minlist} \sqsubseteq \text{est}(R) \Rightarrow \langle \text{listcp}(g_1, g_2) \text{ cat thinlist} \rangle \\
& \text{minlist} \sqsubseteq \frac{\langle [\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}] \rangle}{\supset} \text{est}(R)
\end{aligned} \tag{8.6c}$$

a least-cost bitonic tour, as a fold that thins the tours kept at each city

circuit

$$\begin{aligned}
& \mu F \vdash \frac{1}{\supset} \vdash \text{P}(\mu F) \vdash E(\langle [\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}] \rangle) \vdash \text{P}(\text{Journey}(\text{City})^2) \vdash \text{est}(R) \vdash \text{Journey}(\text{City}) \\
& \text{tour} \vdash \text{est}(R)
\end{aligned}$$

$$\begin{aligned}
& \mu F \vdash \frac{1}{\supset} \vdash \text{P}(\mu F) \vdash E(\langle [\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}] \rangle) \vdash \text{P}(\text{Journey}(\text{City})^2) \vdash \text{est}(R) \vdash \text{Journey}(\text{City}) \\
& = \langle [\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}] \rangle \text{ est}(R) \\
& \langle [\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}] \rangle \triangleq \langle [\text{start}, \text{dropl}] \sqcup [\text{start}, \text{dropr}] \rangle
\end{aligned}$$

$$\begin{aligned}
& \mu F \vdash \frac{1}{\supset} \vdash \text{P}(\text{list}(\text{Journey}(\text{City})^2)) \vdash \text{listcp} \vdash \text{list}(F(\text{Journey}(\text{City})^2)) \vdash \frac{Q_1}{\supset} \vdash \text{list}(\text{Journey}(\text{City})^2) \vdash \frac{Q_2}{\supset} \vdash \text{list}(\text{Journey}(\text{City})^2) \vdash \text{cat} \vdash \text{thinlist} \vdash \text{list}(\text{Journey}(\text{City})^2) \vdash \text{minlist} \vdash \text{Journey}(\text{City}) \\
& \langle \text{listcp}(g_1, g_2) \text{ cat thinlist}(Q) \rangle \text{ minlist}(R) \\
& \sqsubseteq \text{binary thinning theorem, at } P \triangleq T \text{ with merge } \tau = \text{cat}, Q \text{ from } F(Q) \\
& [\text{start}, \text{dropl}] \sqsubseteq [\text{start}, \text{dropl}] \sqcup Q, F(Q) [\text{start}, \text{dropr}] \sqsubseteq [\text{start}, \text{dropr}] \sqcup Q
\end{aligned}$$

$$\begin{aligned}
& \langle [\text{start wrap}, \text{cpr}(\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{ cat thinlist}(Q) \rangle \text{ minlist}(R) \\
& \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \vdash \\
& = [\text{start}, \text{dropl}](\text{l}(p)) = \text{start}(p), [\text{start}, \text{dropl}](\text{r}(q)) = \text{dropl}(\pi_1(q), \pi_2(q)), [\text{start}, \text{dropr}](\text{l}(p)) = \text{start}(p), \\
& [\text{start}, \text{dropr}](\text{r}(q)) = \text{dropr}(\pi_1(q), \pi_2(q)); \text{quadratic, two tours added per step}
\end{aligned}$$

Hinze–Marsden

